

The Evolution of Machine Learning and AI

5 ECTS | Lecturer: Indrakshi Dey

- Department of Computing and Mathematics
- Intermediate level

Why this module matters

AI and ML are reshaping software, decision-making, and digital services.

- symbolic AI → machine learning → generative AI
- expert systems → neural networks → transformers
- narrow tools → intelligent systems

Historical timeline

- 1950s-60s: symbolic AI and early reasoning systems
- 1970s-90s: expert systems and AI winters
- 2000s: statistical learning and data-driven models
- 2010s: deep learning, CNNs, RNNs, and GPUs
- 2020s: transformers, LLMs, and generative AI

Core themes

- classical ML foundations
- neural network training and optimisation
- deep learning families and architectures
- LLMs and software engineering workflows
- ethics, bias, accountability, and transparency

Learning outcomes

By the end of the module, students will be able to:

1. Trace AI and ML milestones and challenges.
2. Compare major model families and training methods.
3. Evaluate modern LLM architectures and their significance.
4. Analyse ethical and practical implications of AI systems.

Assessment and practice

- Continuous assessment: 100%
- Assignment 1: history, concepts, tools, ethics
- Assignment 2: LLMs, workflows, and future trends
- lectures, labs, and applied case studies

Looking ahead

AI is not only about capability; it is also about trust, transparency, and responsible design.